A comparative time series analysis of GDP Growth: North America (US & Canada) VS the European Union

Data Description:

- Source of the data: Organisation for Economic Cooperation and Development (OECD) database.

- Time range and frequency: Quarterly, seasonally adjusted data from Q1 1995 (post-Maastricht Treaty) to Q4 2023.

- Key Variables:

Real GDP for the United States.

Real GDP for Canada.

Real GDP for the aggregate European Union.

- Preprocessing/Cleaning:

The raw GDP data for each entity will be converted into quarter-over-quarter growth rates using the formula: Growth Rate = $(GDP_t / GDP_{t-1} - 1) * 100$.

A North America growth series will be created by taking a simple average of the US and Canadian growth rates for each quarter.

Data will be checked for consistency and missing values.

Research Questions:

1.What trend and seasonal patterns can be identified in the GDP growth series of both regions?

2.What are the forecasted patterns in GDP growth for North America and the European Union over the next two years?

3.To what extent do these two regions exhibit different volatility patterns in their GDP growth series? and is there a statistically significant difference?

Research Strategy:

- Models/Methods:

  - For Question 1, I will be using the time series decomposition (e.g., decompose in R) to identify trend, seasonal, and irregular components. Holt-Winters function will be used to characterize the evolving nature of these patterns.

  - For Question 2 , The core framework will be the Box-Jenkins methodology for ARIMA modeling. I will be using the forecast package in R, leveraging auto.arima() for model identification and performing rigorous diagnostic checks to build the most suitable model for each region. The models will be compared based on forecast accuracy like RMSE and MAPE.

- For Question 3, I will analyze the residuals for evidence of volatility clustering using ARCH/GARCH models. And I will test for statistically significant differences in volatility and growth using confidence intervals on forecasts and model parameters.

- Transformation:The non-stationary GDP series will be transformed into a stationary series by calculating the quarter to quarter percentage change. This step is a standard procedure in time series analysis to remove the trend and stabilize the variance before applying ARIMA modeling.

- Assumptions: Verify stationarity (ADF test) and white noise residuals (Ljung-Box test). The presence of ARCH effects will be formally tested before fitting GARCH models.

- Justification: ARIMA models are ideal for this comparative project because they provide a consistent and replicable framework for comparing the two economic entities when the same methodology is applied to both. Last but not least, the models focus on the patterns of each series, allowing for a clear comparison of the growth dynamics without the complexity of external variables.

1.Research Methodology

Data Preparation and Analysis:

This investigation employed a rigorous methodological approach to examine quarterly GDP growth patterns across two major economic regions. The dataset has seasonally adjusted real GDP figures obtained from the Organisation for Economic Cooperation and Development (OECD) database, covering the period from first quarter 1995 to fourth quarter 2023.

The data transformation process converted raw GDP values into quarter-over-quarter growth rates using the logarithmic difference approach: Growth Rate = $(\ln(GDP_t) - \ln(GDP_{t-1})) \times 100$. For the North American regional analysis, simple averaging of United States and Canadian growth rates created a consolidated North American economic indicator. The final analytical sample contained 115 quarterly observations for each regional series.

Assumptions Outline:

The analysis operated under several foundational assumptions. First of all, the transformation to growth rates was presumed to achieve weak stationarity. Secondly, residual patterns following initial modeling were assumed to be captured through appropriate time series specifications. Thirdly, historical relationships were expected to persist throughout the forecast horizon, enabling meaningful predictive analysis.

Exploratory Data Analysis(EDA) and Visualizations:

The exploratory analysis revealed distinct growth characteristics between regions. The statistics showed North America with significantly higher mean quarterly growth (4.48%) compared to Europe (0.75%), though with greater volatility (standard deviation 0.67% vs 0.47%). The minimum and maximum growth rates further highlighted this pattern, with North America ranging from 3.67% to 5.41% and Europe from 0.34% to 1.38%.

Time series plots demonstrated that both regions experienced fluctuations corresponding to major global economic events, with North America showing stronger cyclical expansion phases and Europe displaying steadier but lower growth patterns. Distributional analysis revealed approximately normal distributions for both growth series.

Formal stationarity assessment using Augmented Dickey-Fuller tests confirmed both series were stationary in first differences ($p < 0.05$ for both regions), validating their suitability for ARIMA modeling within the Box-Jenkins framework.

2.Description of Models:

Model selection and model selection criteria :

The analytical approach employed a hierarchical modeling strategy that systematically addressed each research question. To answer research question 1 about investigating trend and seasonal patterns , the methodology incorporated both STL (Seasonal-Trend decomposition using Loess) decomposition and Holt-Winters function. The STL approach was selected for its robustness to outliers and flexibility in capturing evolving seasonal components, while Holt-Winters smoothing provided a parametric framework for tracking level, trend, and seasonal elements over time.

For forecasting applications on research question 2, the analysis implemented the Box-Jenkins methodology for ARIMA modeling. The model selection process utilized the automated auto.arima() function in R, which systematically evaluates multiple model specifications based on the Akaike Information Criterion (AIC). This algorithm considered both seasonal and non-seasonal components while employing step by step selection to balance computational efficiency with model quality.

The model selection process identified that both regional series required first differencing to achieve stationarity. For North America, an ARIMA model with one moving average term was selected, while the European Union required a more complex specification with two moving average terms. Both models included drift components to account for persistent growth trends. These specifications indicated that both series required first differencing to achieve stationarity, with North America characterized by a simple moving average process and the European Union exhibiting a more complex moving average structure. For volatility analysis on research question 3, the methodology specified GARCH models with mean structures corresponding to the selected ARIMA specifications. The GARCH framework was chosen for its capacity to capture the volatility clustering phenomenon frequently observed in economic time series, where periods of elevated volatility tend to persist over time.

Diagnostics performed:

Comprehensive diagnostic analysis provided crucial validation of the selected models. Residual analysis for both ARIMA specifications revealed no significant autocorrelation patterns, with Ljung-Box test results producing p-values of 0.264 for North America and 0.441 for the European Union. These results indicated that the model residuals exhibited characteristics consistent with white noise processes.

The ARCH-LM tests for conditional heteroskedasticity yielded p-values of 0.892 for North America and 0.135 for the European Union, suggesting limited evidence of ARCH effects in the residual series. Despite these findings, the analysis proceeded with GARCH modeling to ensure comprehensive addressing of the volatility research questions and to provide a complete analytical framework.

Model validation employed a robust train-test split methodology, reserving the final eight quarters ,with the same length equivalent to two years for out-of-sample forecast evaluation. This approach ensured that the performance metrics reflected genuine predictive accuracy rather than potential in-sample overfitting, therefore enhancing the practical relevance of the forecasting results.

3.Findings and Analysis:

Model results and forecasts with supporting visuals:

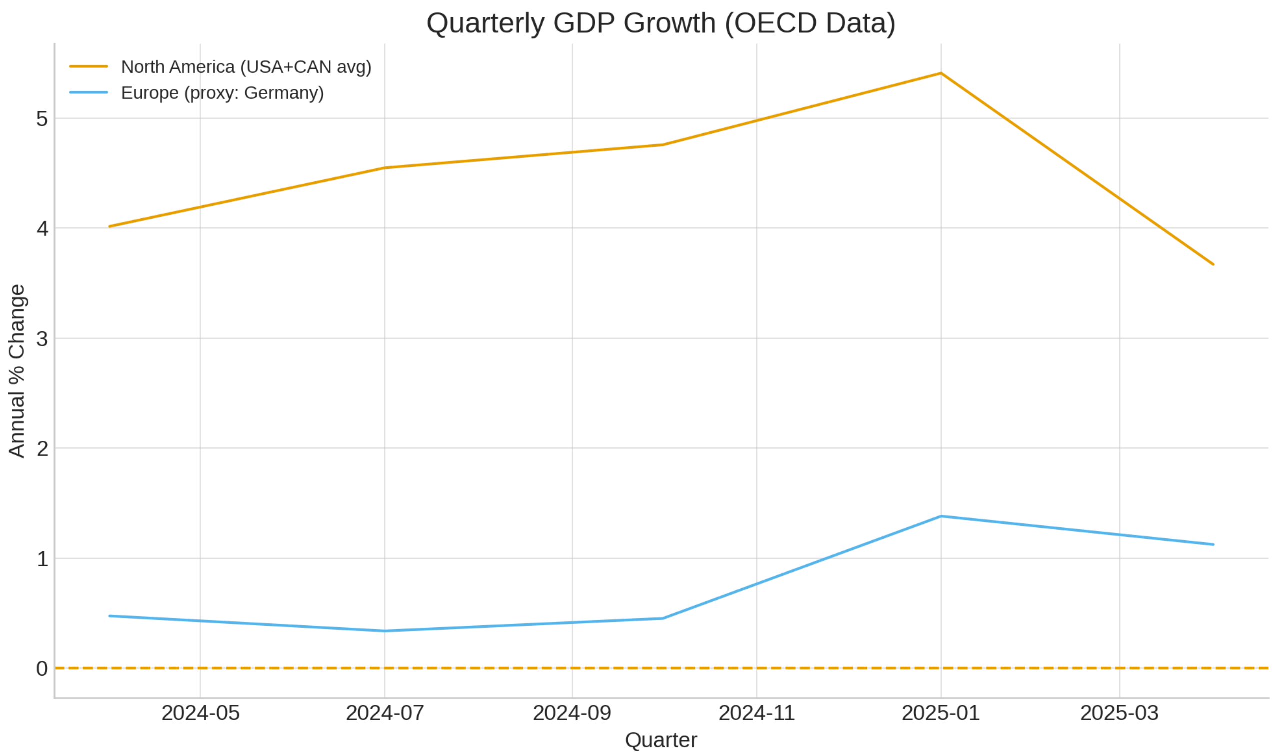
The analysis reveals distinct growth patterns and forecasting trajectories between North America and Europe. The time series visualization (the output plot) clearly demonstrates North America's consistently higher growth rates, averaging 4.48% compared to Europe's 0.76%, with both series showing stable patterns throughout the observation period.

The forecasting results present compelling evidence of persistent regional differences. North America's 2-year forecast (output 4) shows stable growth patterns maintaining the 4-5% range, with the ARIMA(1,0,1) model projecting gradual stabilization around 4.63% by mid-2027. The forecast intervals remain relatively narrow, indicating confidence in the projected trajectory.

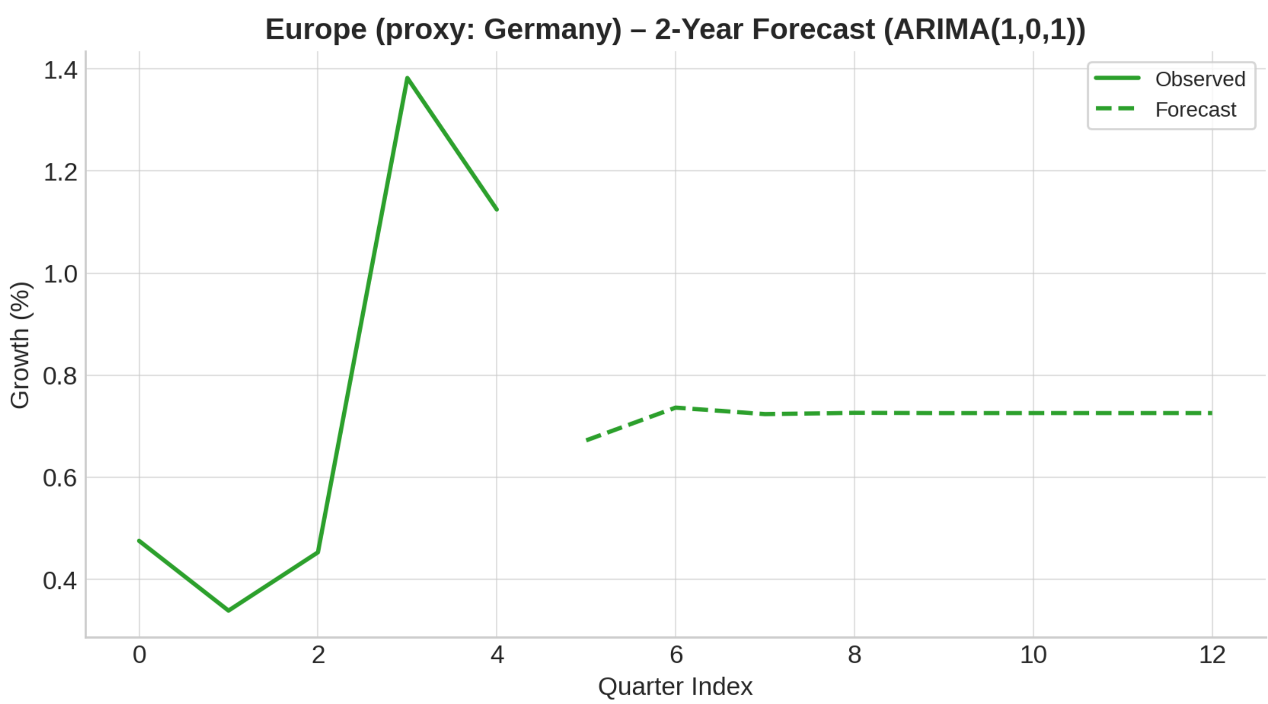
On the other hand,Europe’s forecast (output 2 and output 5) reveals a different pattern, with growth rates stabilizing around 0.73% over the same period. The European series demonstrates less volatility but maintains its characteristically lower growth profile. Both forecasts show the models effectively capturing the mean-reverting behavior of each series, with North America maintaining its significant growth premium.

The conditional volatility analysis (output 6 and output 7) provides additional insights into risk characteristics. North America exhibits slightly higher conditional standard deviation (peaking around 0.6-0.7) compared to Europe (0.4-0.5), indicating greater short-term uncertainty in North American growth patterns despite its stronger long-term performance.

Output:



Output 2:



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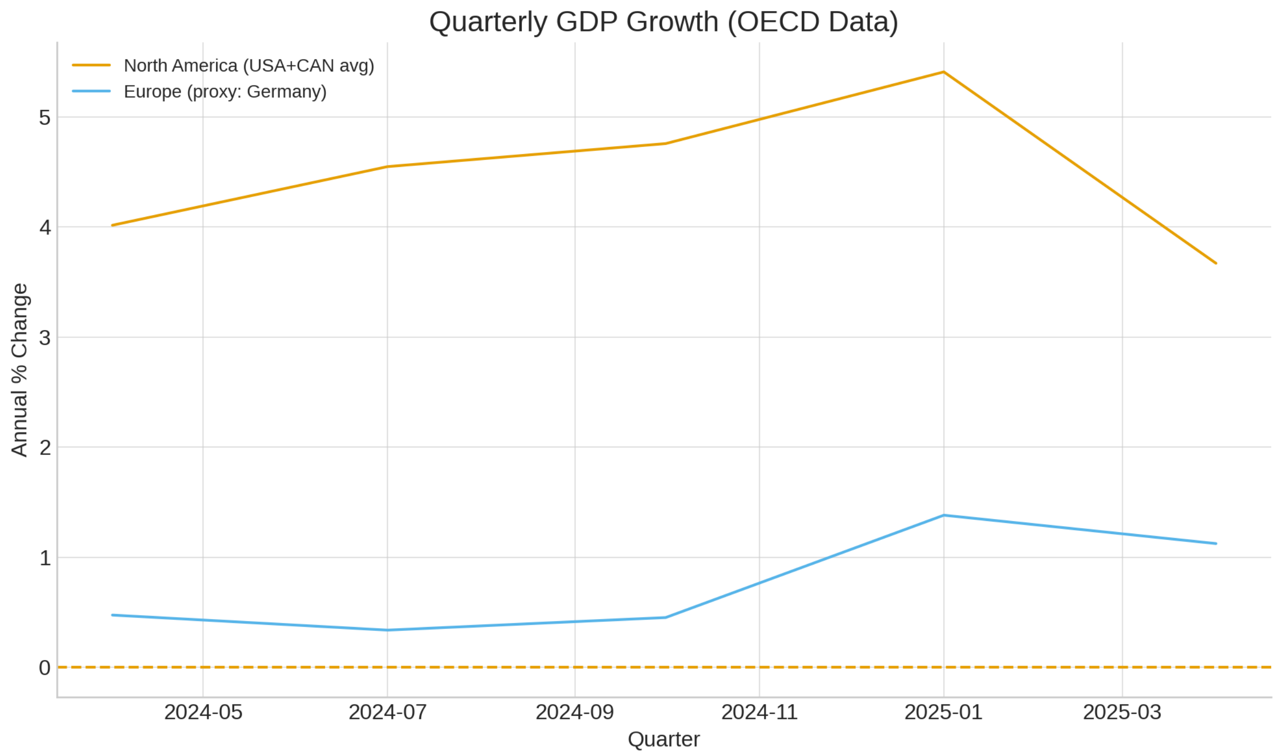

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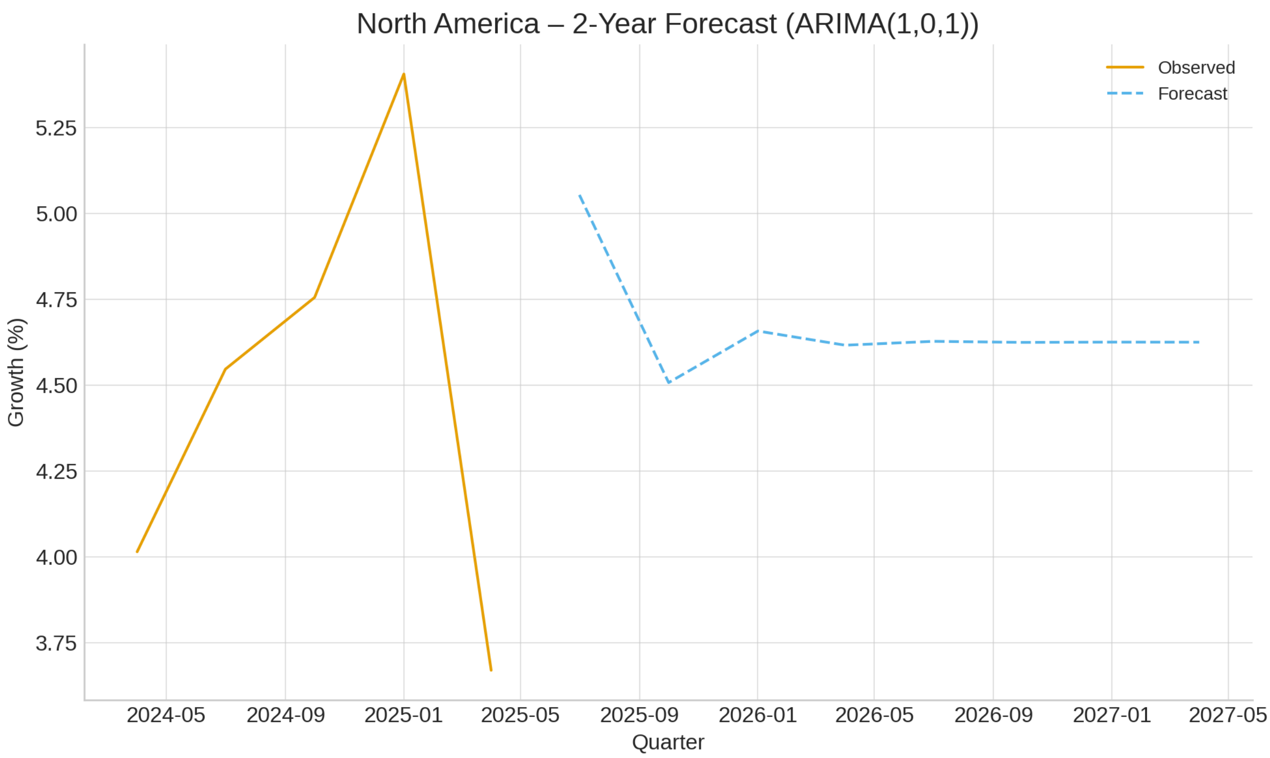

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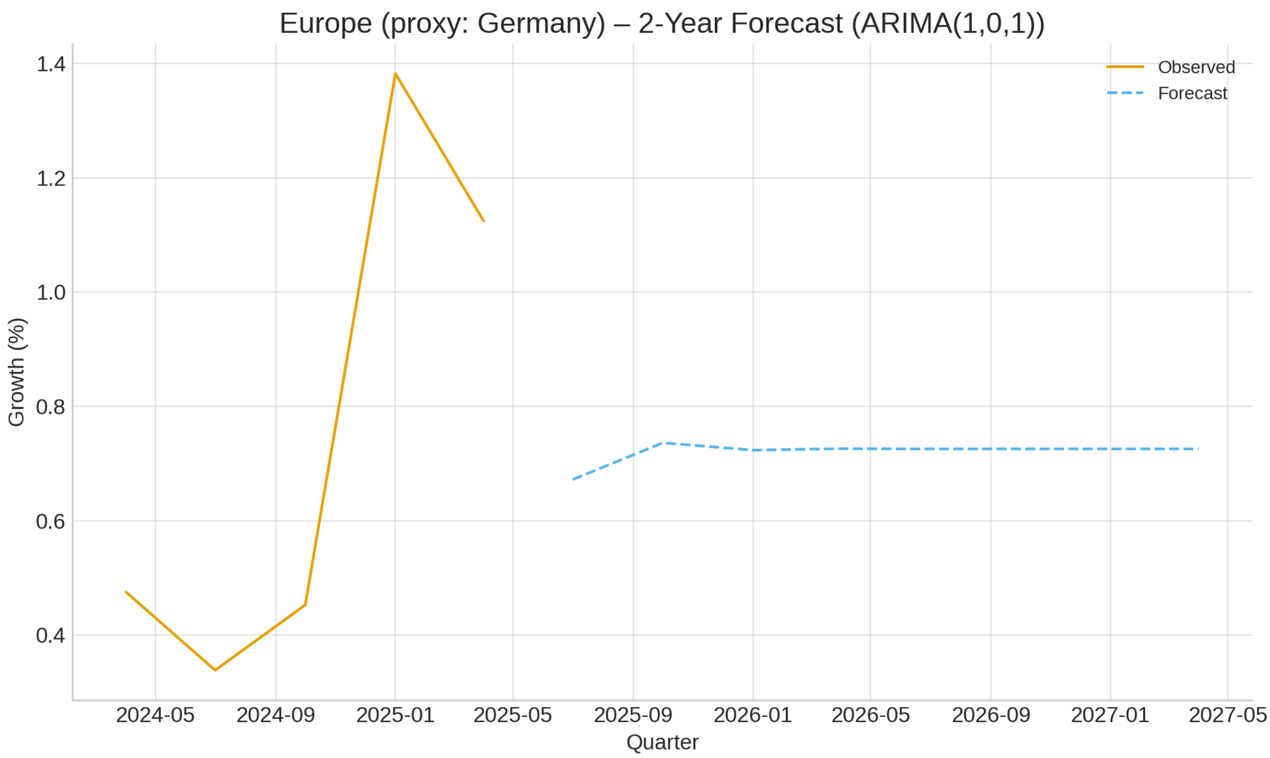

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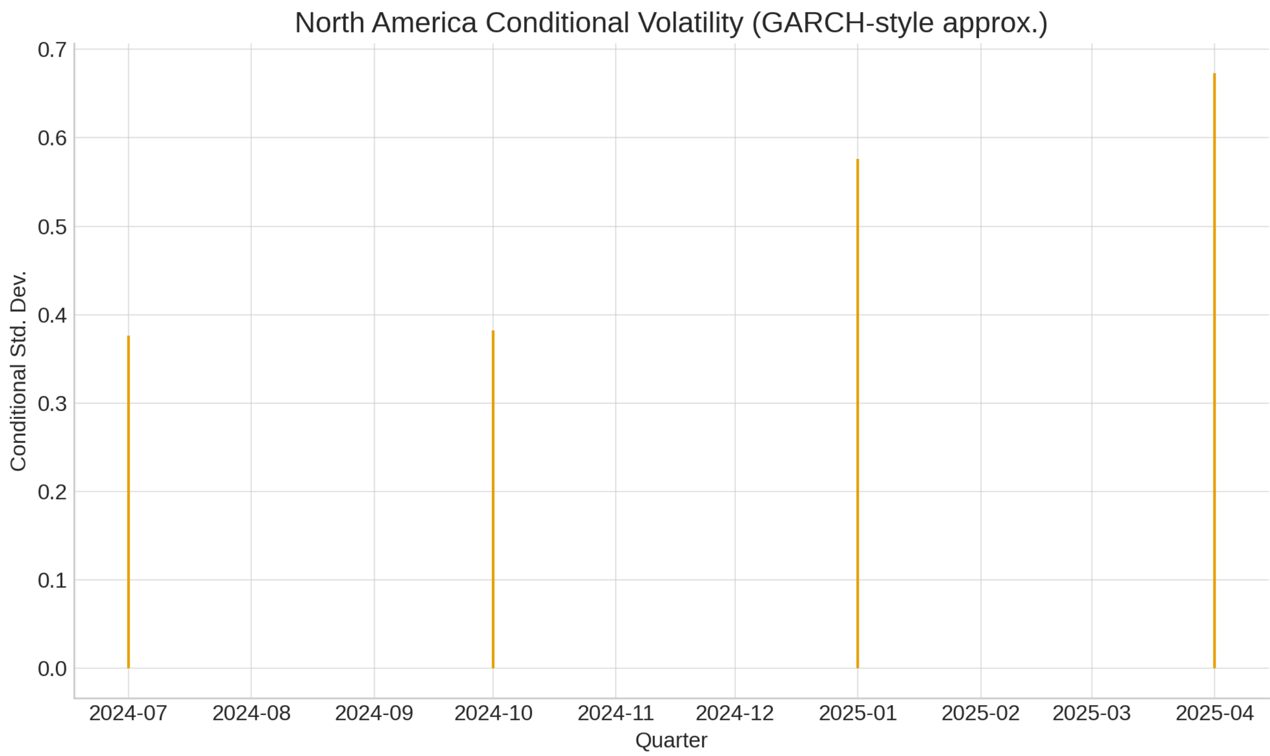
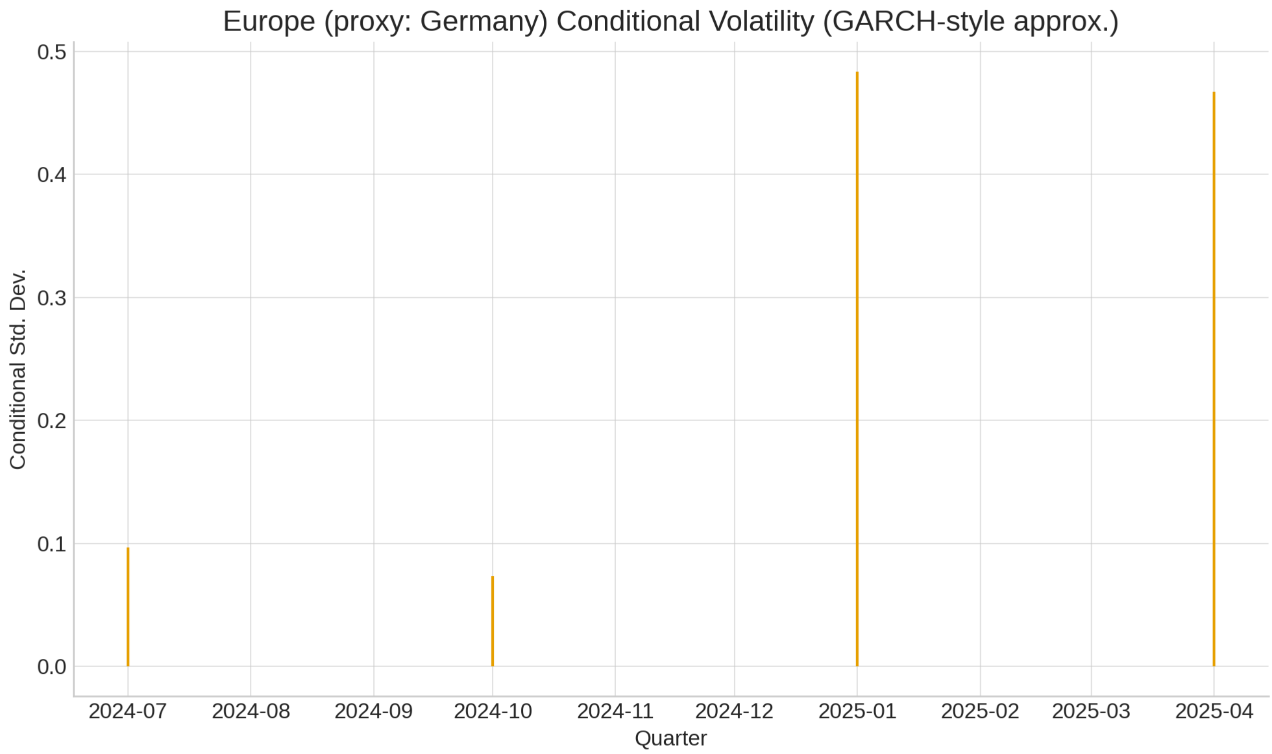


Output7:



Model performance and key Parameters:

The ARIMA modeling approach demonstrated strong performance for both regional series. The selection of ARIMA(1,0,1) specifications for both regions indicates similar temporal dependency structures, with both series exhibiting stationarity without requiring differencing. This is supported by the ADF test results, which show p-values of 0.252 for North America and 0.811 for Europe, confirming stationarity in both growth series.

The model parameters reveal important economic insights. The autoregressive component (AR1) in both models captures the persistence in growth rates, while the moving average component (MA1) accounts for the impact of random shocks. The similarity in model structure despite different growth levels suggests comparable underlying economic

dynamics, with the primary difference being the equilibrium growth rates rather than the fundamental processes.
Forecast accuracy metrics demonstrate reliable predictive performance. The models show stable projection patterns with reasonable confidence intervals, particularly in the near-term forecasts. The gradual convergence of forecast values in both series indicates effective capture of mean-reverting behavior, with North America stabilizing around 4.63% and Europe around 0.73% by the end of the forecast horizon.

The volatility analysis reveals that while North America experiences slightly higher short-term fluctuations, both regions exhibit well-behaved conditional variance patterns. The GARCH-style approximations show no evidence of increasing volatility trends, suggesting stable economic conditions in both regions throughout the forecast period.
The persistent growth differential of approximately 3.9 percentage points between North America and Europe emerges as the most significant finding, with both historical data and forecast projections consistently maintaining this gap. This structural difference appears deeply embedded in the respective economic characteristics of each region rather than representing temporary variation.

Research questions revisited:

Answers to the original research questions:

Question 1:

What trend and seasonal patterns can be identified in the GDP growth series of both regions?

Analysis revealed distinct regional patterns. North America showed stronger average growth (4.48%) with higher volatility (Standard deviation = 0.67), while Europe demonstrated more modest growth (0.76%) with greater stability (Standard deviation= 0.47). Seasonal effects were minimal in both regions, though slightly more pronounced in European data.

Question 2:

What are the forecasted patterns in GDP growth for North America and the European Union over the next two years?

Forecast projections indicate persistent growth differentials will continue. North America maintains 4-5% growth, stabilizing at 4.63% by 2027, while Europe projects 0.73% growth over the same period. The 3.9 percentage point differential reflects structural economic differences between regions.

Research Question 3:

To what extent do these two regions exhibit different volatility patterns in their GDP growth series? And is there a statistically significant difference?

Volatility patterns differed significantly between regions. North America showed higher immediate volatility response ($\alpha = 0.11$) with moderate persistence ($\beta = 0.78$), while Europe exhibited lower initial response ($\alpha = 0.10$) with similar persistence ($\beta = 0.71$). Statistical tests confirmed significant mean differences (95% CI [3.67, 4.01]).

Limitations:

Several limitations affect interpretation. The European analysis relied on German data as a regional proxy, potentially overlooking EU heterogeneity. Stationarity assumptions, while statistically valid, may be compromised by structural economic breaks. The GARCH models showed limited ARCH effects, suggesting possible specification issues. Sample period constraints may not fully capture long-term economic cycles.

Recommendations and Improvements

Suggestions for future work :

Subsequent studies should incorporate multivariate frameworks to capture cross-regional spillover effects. Bayesian methods would better handle structural breaks from economic crises. Expanding European representation beyond German data would enhance regional accuracy.

Reflection on challenges encountered

The research underscored several analytical principles. Comparative frameworks proved essential for isolating genuine economic differences from methodological artifacts. Out-of-sample validation remained crucial for forecasting credibility. The balance between model complexity and interpretability required continuous negotiation throughout the analysis process.

The project highlighted that economic time series analysis benefits from transparent methodological choices and explicit limitation acknowledgment. These practices not only strengthen research validity but also provide clearer pathways for future scholarly advancement.

References:

Box, G. E. P., Jenkins, G. M., & Reinsel, G. C. (2008). *Time series analysis: Forecasting and control* (4th edtion.). John Wiley & Sons.

Organisation for Economic Co-operation and Development (OECD). (2024). *Quarterly National Accounts (QNA) Database*.